

Habitability is bigger than one habitable zone

- **Stellar:** liquid water needs the right distance from a star for long enough.
- **Galactic:** our Milky Way neighborhood is safer than the crowded center but still metal-rich enough for rocky planets.
- **Planetary:** the carbonate-silicate cycle can buffer CO₂ and climate over geologic time.
- **Moons:** giant planets are harsh, but their moons can share the star's habitable zone and gain tidal energy.
- **Time:** if observers can appear across vast future ages, our early arrival needs explanation.

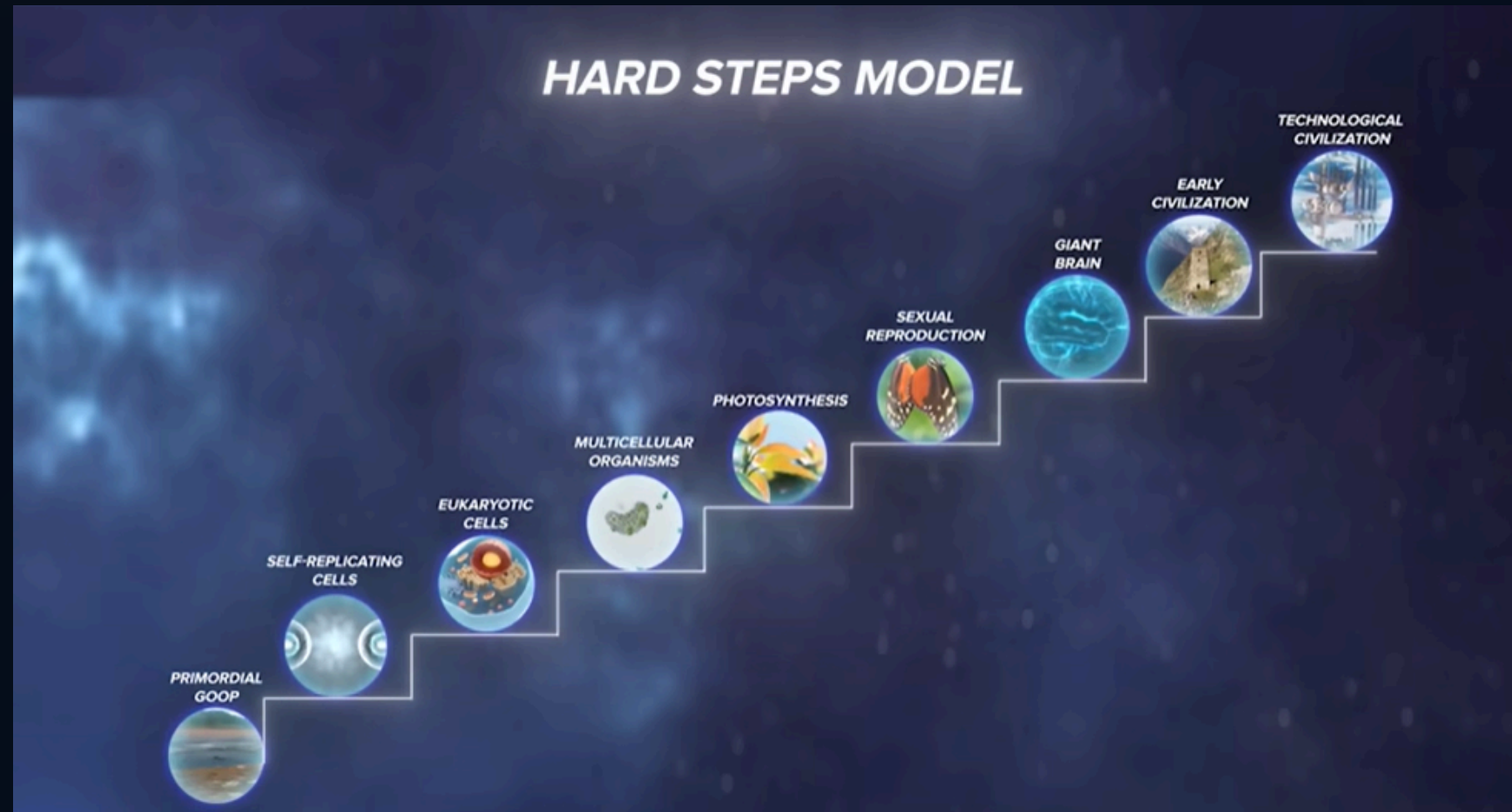


STELLAR HABITABLE ZONE

16.8 · FILTERS ACROSS HARD STEPS

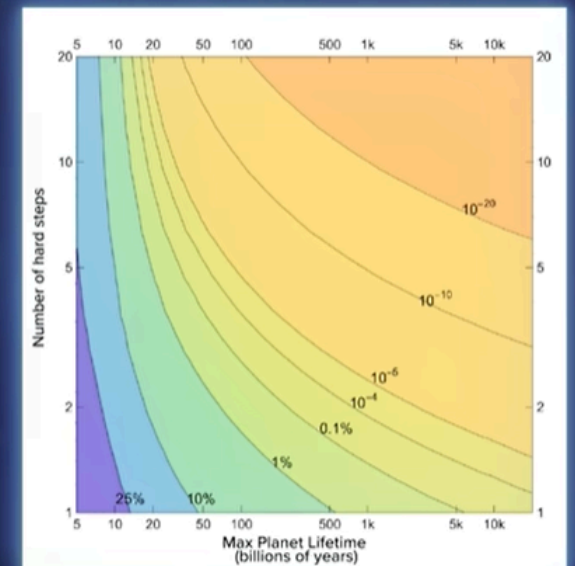
The timeline question is a filter question

Copernican: do not assume our place is special. **Anthropic**: we can only find ourselves in a moment that supports observers. The **hard-steps model** turns that tension into a chain of fluke events — rare per year, but inevitable given enough time.



Hard-steps staircase · goop → self-replicating cells → eukaryotes → multicellularity → photosynthesis → sex → brains → civilization

Both visuals are from "If Loud Aliens Explain Human Earliness, Quiet Aliens Are Also Rare" — Hanson, Martin, McCarter, and Paulson, 2021.



FROM HANSON, MARTIN, MCCARTER & PAULSON (2021)

Birth-rank surface: are we typical, or wildly early?

Before LUCA, chemistry had to become evolvable

LUCA

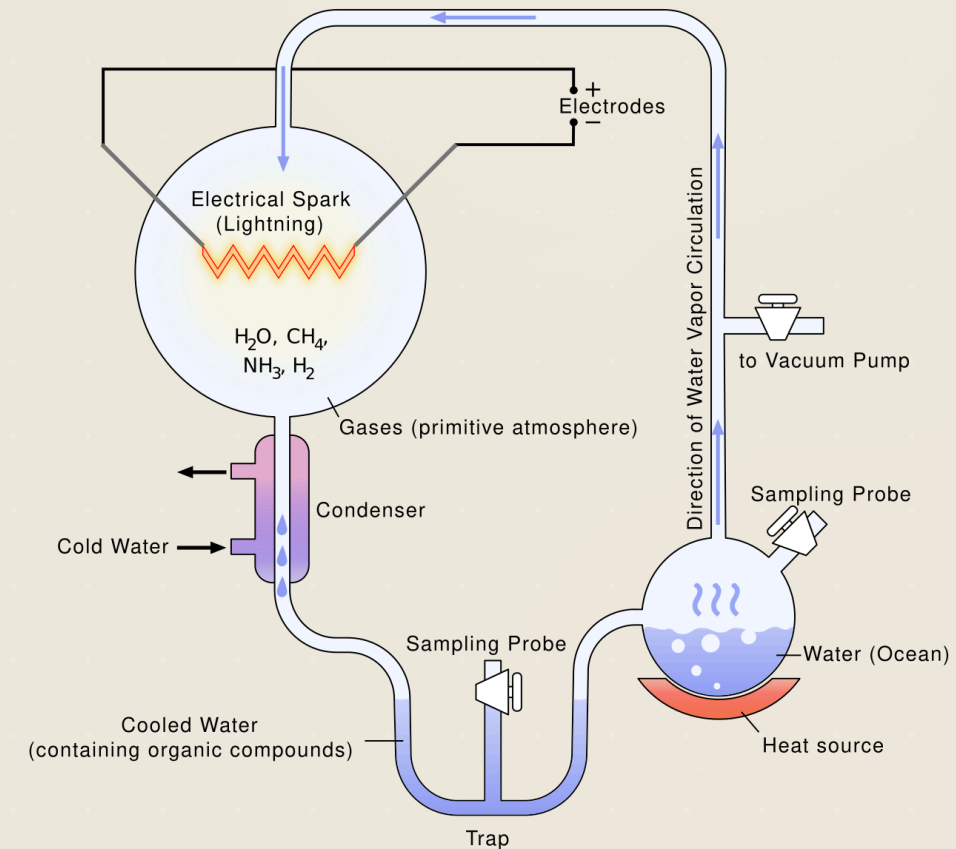
Last Universal Common Ancestor — not the first living thing, the most recent ancestor shared by all known life today.

Clays matter

Open ocean is too dilute for long polymers. Clay surfaces concentrate, align, and catalyze the long chains RNA and peptides need.

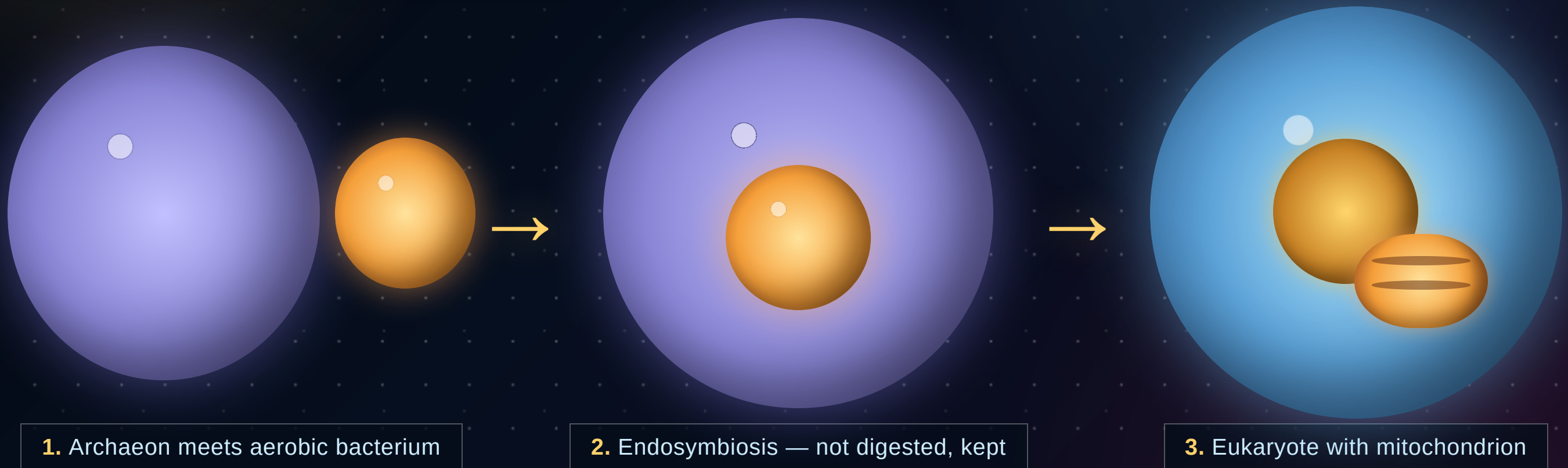
Asymmetry hook

CERN's heavy-neutron experiment broke matter-antimatter symmetry in ~2.5% of ~80,000 decays — tiny biases can decide what gets to exist.



Miller-Urey apparatus, 1952 — simulated atmosphere, an electric spark for "lightning," and a small ocean. Result: amino acids in the trap.

Eukaryogenesis: a once-only merger that made complexity possible



Around 2 billion years ago, a stressed-out archaeon absorbed a bacterium and kept it. The bacterium became **mitochondria**; the cell-size ceiling broke; complex life became possible. Simple life appeared early, extinctions never finished it, multicellularity evolved more than once — this absorption seems to have happened only once.